

Size and Momentum Profitability in International Stock Markets

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This version: February 8, 2017

Abstract

We study the link between the profitability of momentum strategies and firm size, drawing on an extensive dataset covering 14 stock markets across the globe. International momentum profitability is markedly higher in medium-size than in big stocks. Momentum premia are considerably diminished by trading costs, when taking into account the actual portfolio turnover incurred when implementing this strategy. Thus, while small stocks exhibit momentum, exploiting this feature is hardly feasible. By contrast, momentum premia especially for medium-sized stocks still remain economically and statistically significant in most equity markets worldwide after adjusting for trading costs.

JEL classification: C89, G12, G15

Keywords: Size; momentum; international equity markets; asset pricing anomalies; trading costs

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1. Introduction

This paper investigates the interplay between size and momentum strategies in international stock markets. The main focus of our work is to determine if any momentum profits are indeed realizable in the practical implementation of investment strategies or whether they have to be deemed illusory. Our analysis of international data is motivated by the fact that researchers and practitioners alike are increasingly eager to determine the existence or non-existence of equity market anomalies across the globe. Also, relying on international data can help address the concern that anomalies observed in the US market alone may possibly be a manifestation of survivorship or data-snooping biases (Kothari et al., 1995; Lo and MacKinlay, 1990; MacKinlay, 1995). Moreover, using a cross-section of markets allows for an analysis of the common roots of a specific anomaly.

Momentum has long been recognized to be an important factor when modelling expected returns. Jegadeesh and Titman (1993) show for the US that stocks having performed well in the past twelve months perform significantly better in the next 3-12 months than stocks which have performed poorly in the past twelve months. The existence of momentum profits has also been documented in international data (e.g. Chui et al., 2010; Rouwenhorst, 1998), though it appears to be surprisingly weak in emerging markets (Cakici et al., 2016).

The interplay of size and momentum, by contrast, has received rather little attention so far. This is surprising given the enormous importance of the momentum strategy in practice and the corresponding need for insight into whether momentum profits are available in all size categories of stocks. Theoretical considerations suggest that this need not, in fact, be the case. The gradual information flow hypothesis of Hong and Stein (1999) holds that information spreads with different speeds for different asset classes. Asset classes with slower diffusion of (firm-specific) information exhibit more pronounced momentum. Since analyst coverage and firm size are highly correlated, one

would therefore expect market imbalances to adjust more quickly for bigger stocks than for smaller ones, implying reduced or no momentum for large-cap stocks. Conversely, we consider the notion that although small-cap stocks appear most promising as a source of momentum profits, trading costs and lack of market depth may render momentum strategies for these stocks infeasible.

[Table 1 here]

Table 1 provides an overview of the existing studies on this question and positions this paper in the literature. The major studies we are aware of that address the interplay of size and momentum are Jegadeesh and Titman (1993), Hong et al. (2000), Jegadeesh and Titman (2001), Fama and French (2008), Avramov et al. (2013), and Israel and Moskowitz (2013).¹ All these studies use US data, but the evidence on the size-momentum relationship is mixed. Jegadeesh and Titman (1993) form three size groups and examine whether the profitability of momentum strategies is confined to particular size groups. They find that this strategy works for all three size subgroups. Hong et al. (2000), with the specific intent to test the Hong and Stein (1999) hypothesis, sort stocks on size and momentum and find for US data from 1980 to 1996 that “... once one moves past the very smallest capitalization stocks (...) the profitability of momentum strategies declines sharply with market capitalization” (p. 267). Jegadeesh and Titman (2001) also sort by size and momentum and find a positive and significant momentum spread in every size group. However, they only use two (small and big) size groups, whereas in our paper most of the action is found in the middle size group. Fama and French (2008) detect a significant and positive momentum spread for all three size groups they examine. Avramov et al. (2013) find a significant momentum spread for all three size groups they

¹ A number of other studies include results that indirectly speak to the question of this paper. For example, Korajczyk and Sadka (2004) examine the impact of portfolio size on momentum strategies with US data. Novy-Marx and Velikov (2016) present results for US data for momentum portfolios by size.

study. Israel and Moskowitz (2013) examine the momentum spreads of each of the five size groups from the 25 portfolios provided by Kenneth French. They observe that “momentum premia are present in every size group” and, in some contrast to Hong et al. (2000), “find no consistent evidence that momentum returns decrease with firm size” (p. 295). All six studies use CRSP data for the US, though they differ widely in terms of the time period covered (see Table 1), which may explain some of the differences in results.

This paper uses an international dataset for a broad range of countries (including the US), covering, in general, 1991 to 2012. The first contribution of our analysis, therefore, is to provide an out-of-sample analysis of the relationship of size and momentum, in terms of the time span, country coverage, and data source.² A related study, using international data on the country level is Cakici and Tan (2014). They also examine the momentum returns of different size groups, like we do. However, since they only use two size groups, they do not examine a medium size group, where we detect the most profitable momentum returns. They also do not account for trading costs and do not use a factor model. Different from the present study, Fama and French (2012) examine regional, not country-level, data, and do not account for trading costs (which we discuss below). Their study implies positive momentum spread returns for the two smallest of five size groups, but do not test for the significance of these returns.³ We find similar results using country-level data, documenting statistically significant momentum spread returns mostly for medium-size stocks, but also, to a smaller extent for big-size stocks and, before adjustment for trading costs, also in small-cap stocks. Similar to known evidence from the US (Avramov et al. (2013)), we find that for international momentum strategies, the ability to short loser stocks plays an important role. We do however, also

² Chui et al. (2010) show that momentum profits in a cross-section of countries are negatively related to the median firm size in the respective markets.

³ Fama and French (2012) report excess returns and t-statistics of the 25 portfolios sorted on size and momentum, but no excess returns of the spread between winners and losers. While the excess spread returns can be easily calculated using their results, a calculation of the t-statistics is therefore not possible.

find that the long side considerably adds to the momentum premium.

In sum, our first contribution is to confirm, out-of-sample, the observation previously made by Hong et al. (2000) for US markets that the profitability of momentum strategies is higher for medium-sized stocks than for big stocks.

Our second contribution is to incorporate trading costs in various ways into the analysis, therefore critically assessing how feasible it is to practically implement investment strategies based on size and momentum. Almost all listed studies use at least a simple model for risk correction. However, as seen in Table 1, most existing studies account for trading costs only in a simple way, or do not consider trading costs at all. Nonetheless, trading costs can be an important consideration in practice and are receiving attention in the current literature (e.g. Frazzini et al., 2012; Novy-Marx and Velikov, 2016). We employ three complementary approaches. First, we directly approximate trading costs for each portfolio by using fixed values per transaction combined with the actual portfolio turnover. Second, we compute critical (maximum) trading costs that an investor could occur and still earn significant excess returns. Third, we calculate the US dollar trading volume of the involved portfolios to assess whether the market has enough depth and capacity for such a quantitative strategy to be implementable in practice.

We find that the theoretically significant momentum premia of small-cap stocks are much reduced when taking into account trading costs. However, for most countries momentum premia for medium-sized stocks, and also for big-sized stocks, but to a smaller extent, remain statistically and economically significant.

We discuss data selection and preparation issues in a companion paper (Schmidt et al., 2017). There we compare US market returns and common risk factors from our dataset, compiled with Thomson Reuters Datastream (TRD) and Thomson Reuters Worldscope (TRW) data, with the corresponding series from Kenneth French's website. To delete data errors as well as extreme

outliers, we employ specific screening procedures.⁴ We are able to replicate very closely the properties of benchmark data, suggesting that the screens are effective in transforming the raw data into a data series suitable for further analysis.

The paper proceeds as follows. Section 2 presents the empirical baseline results for momentum strategies conditional on size. Section 3 includes trading cost measures. Section 4 concludes.

2. Size and profitability of momentum

In this section, we consider the interrelation of size and momentum.

2.1. *International size and momentum portfolios*

2.1.1 *Momentum conditional on size*

We sort by size and momentum jointly (three-by-three) and construct long-short portfolios for each size group.⁵ We present results for raw returns as well as three-factor alphas of winner-loser momentum spread portfolios to judge whether there is a momentum effect conditional on size. The three-factor alphas are estimated by considering the intercepts of a regression of the spread portfolio on the market, as well as the SMB and HML factors.⁶

$$(r_t^{w,g} - r_t^{l,g}) = \alpha + \beta_M * (r_{M,t} - r_{f,t}) + \beta_{SMB} * SMB_t + \beta_{HML} * HML_t + \varepsilon_t \quad (1)$$

where $r_t^{w,g}$ ($r_t^{l,g}$) is the return of the winner (loser) tercile of size group g in time t ,⁷ $r_{M,t}$ is the return

⁴ For the data preparation process, we employ static and dynamic screens as suggested by Ince and Porter (2006) as well as additional filters.

⁵ This is an independent sorting scheme, and not a sequential one.

⁶ Results with the Carhart (1997) four-factor model are available on request and generally similar. However, since the test and factor portfolios are constructed similarly, the momentum factor might be a bad control variable.

⁷ Like Israel and Moskowitz, (2013), we use the previous 12 months with skipping the most recent one for predicting the returns of the next month, like we did with the construction of the momentum factor in Appendix A. (Jegadeesh and Titman (1993) use a six-month/six-month strategy.) The main reason for using the 12 month version of momentum is to

of the market portfolio in time t , $r_{f,t}$ is the riskfree rate proxy in time t , SMB_t , and HML_t are the factors as described in Appendix A, β_M , β_{SMB} , and β_{HML} are the corresponding factor loadings, α is the three factor alpha, and ε_t is an error term.

[Table 2 here]

The results are shown in Table 2. A monotonic relationship between size and momentum where momentum premia decrease with size is only observable for Italy and the Netherlands. However, the returns of the middle size group are bigger than the returns of the big size group for twelve out of 14 countries (see column (7) of Table 2). For the US, we reject the null hypothesis of no difference between the middle long-short portfolio and the big long-short portfolio returns with a t-statistic of 2.06 at the 5% significance level (see column (8)). In total for seven (five) countries the difference of the returns is positive and significant at the 10% (5%) level.⁸ The countries for which the differences are significant at the 5% level include major markets like the US, UK, Germany, and France.

One of the two countries where we do not find the same result is Japan, a known exception. The other one has a rather small stock market (Norway). These results do not seem to be particular to our specific time period and the result is qualitatively the same if we consider different time periods (results are not reported, but are available on request). Overall, these international results are in line with the US results of Hong et al. (2000).

ensure comparability with the factors provided by Kenneth French.

⁸ These results are similar to the results of Israel and Moskowitz (2013), but differ in conclusion. Israel and Moskowitz (2013) split the long-short momentum returns into five groups, from small to big stocks. They consider CAPM alphas and test if the difference of the smallest and the biggest long-short portfolio returns is significant. They cannot reject the null hypothesis for the US with a t-statistic of 1.56. We instead split the long-short momentum returns into three groups, from small to big stocks. Here, we use raw returns (four-factor alphas are discussed in the following) and test if the difference of the middle long-short portfolio and the big long-short portfolio returns is significant. Given the differences in the considered sample period and methodology we consider the difference in the t-statistic as rather small (1.56 vs. 2.06), though the conclusion of the two studies is different.

The momentum spreads of small stocks are for most countries either similar to those of medium stocks or smaller. Only for one country (Netherlands) are momentum spreads for small stocks larger than for medium-size stocks. These results are also in line with Hong et al. (2000). Since this result is reinforced by the following analyses and profits for small stocks are much harder to realize, we focus in the following mostly on the results of medium and large stocks.

The results of the momentum premia conditional on size for the estimated three-factor alphas are shown in Table 3. Again, the momentum premium decreases (sharply) in almost all countries (with Japan and Norway as exceptions) when moving from medium size to big size stocks, as shown in column (7). Moreover, for seven (two) of these countries this difference is significant at the 10 % (5%) level.

[Table 3 here]

In sum, we consider these findings on international stocks as (modest) evidence in favor of the prediction of Hong et al. (2000). Another interpretation of these results is to relate them to the findings of Chui et al. (2010), who suggest that momentum is related to individualism.⁹ The individualism measure of Chui et al. (2010) is indeed highly correlated with our average momentum returns across countries. We get cross-country correlations of the average momentum returns of the small, medium, and big size groups with the static individualism measure according to Hofstede (2001) as high as 0.44, 0.54, and 0.58. The correlation coefficients with the three-factor alphas are similar with 0.38, 0.56, and 0.61. This suggests that individualism may be connected to average momentum returns for middle size and also for big stocks, but to a lesser extent for small stocks.

⁹ Chui et al. (2010) suggest that momentum profits are driven by behavioral biases, like overconfidence or self-attribution. Namely, they relate cross-country cultural differences to momentum. These differences are measured by a static individualism measure proposed by Hofstede (2001). They find that the individualism measure is positively associated with momentum profits.

2.1.2 *The role of shorting*

Avramov et al. (2013) argue that momentum strategies derive their profitability from shorting loser stocks in the first place. They argue that this observation is mainly driven by high credit risk firms that experience deteriorating credit conditions. To examine the role of shorting in our international data, Table 4 reports the raw returns in excess of the risk free rate for the nine portfolios jointly sorted on size and momentum.¹⁰

[Table 4 here]

Interestingly, for the smallest size group the short position (column (1)) never contributes substantially to the momentum premium and often lowers it considerably.¹¹ Always taking the long position in winners for small stocks gains a positive premium, which is always significant, with the exception of Japan. For medium size stocks the results point to no significant contribution of the sort position in loser stocks in both directions. There is only one exception for which the short position contributes significantly to the momentum premium (Germany). Taking always the long position in winners for medium size stocks also leads to a positive premium for all countries. Among those, eleven are significant (5% level). For big stocks the picture is similar as for medium size stocks. The portfolio return of the presumed short position in losers (column (7)) itself is never significantly different from zero and all but one long position for winner stocks is positive and ten are

¹⁰ One may be tempted to compare the results of Table 4 with Table 3 of Hong et al. (2000). However, a comparison is difficult due to three reasons: 1. Hong et al. (2000) sort by momentum first, then by size, whereas we conduct a joint sort. 2. Hong et al. (2000) form ten size subgroups whereas we form three. Simply averaging the three smallest (biggest) size groups and comparing the results with our terciles is not sufficient because this ignores the value weighted impact of the stocks to the overall returns. 3. We use a different time period: Our period ranges from 1991 to 2012, whereas the time period of Hong et al. (2000) ranges from 1980 to 1996.

¹¹ We caution that in this international study we have to employ a three-by-three grouping in order to have a sufficiently large number of stocks in each portfolio. It is possible, in the light of evidence provided by Bandarchuk and Hilscher (2013), that more extreme losers have more extreme low returns also in international data.

significantly different from zero (10% level).

Table 5 presents the results for the estimated risk-adjusted returns. For small stocks the picture is rather mixed. The estimated alphas of the presumed long position in winners are always positive and significantly different from zero (5% level). However, the estimated alphas of the presumed short position in losers are positive (and significant on the 5% level) for six and negative for eight countries (four (three) of them are significant on the 10% (5%) level). For the medium size stocks the presumed short position in loser stocks is negative for all but one country (Japan). Moreover, for eleven out of fourteen countries the short position is significantly different from zero on the 5% level. Winner stocks do also contribute to the momentum premium with a positive mean return for all countries (significant on the 10% level for nine out of fourteen countries). However, the returns of the loser stocks are often higher in absolute value than the returns of the winner stocks. For the big size stocks, we find similar results as for medium size stocks. The alphas of loser stocks are negative for all countries (significant on the 10% level for nine countries). The winner alphas are positive for all but one country (Hong Kong) and significantly so for seven of them (10% level).

In sum, these results yield mixed results on the importance on shorting. For raw returns, our results indicate that shorting abilities are not that important. However, the results with the three-factor model suggest that shorting does play a role, in particular for medium and big size stocks.

[Table 5 here]

2.2 *Summary*

Overall, our results imply a positive and sizeable momentum premium especially for medium size stocks. The medium-size momentum returns are, apart from two exceptions (Japan and

Singapore), for most countries about 1% per month (13% annually), sometimes even higher, up to 1.80% per month for Australia (24% annually). This is also much bigger than the market excess return of the US. Momentum for medium-sized stocks is stronger than for large-cap stocks, confirming out-of-sample the results of Hong et al. (2000).

The question to be addressed now is whether these strategies are implementable once trading costs are taken into account.

3. Size and momentum when adjusting for trading costs

In this section we expand the analysis of Section 2 with a view towards the actual implementability of the strategies. To do so we apply three different, but complementary approaches, described in Section 3.1. The first one assumes fix trading costs per trade and utilizes turnover data to calculate trading costs for each portfolio. The second one does not assume any trading cost a priori, but computes implicit “critical trading” costs that one could bear and still obtain marginally significant results with the assumed trading strategies. Third, we calculate the monthly trading volume of the portfolios examined, to assess whether it is possible to realize such strategies without too much price impact (which would drive the trading cost beyond the numbers assumed for the two approaches described above). Section 3.2 then reports the results.

3.1. Trading costs and trading volume

In our first approach, we utilize portfolio turnover data, positing a fixed amount of costs per trade. We use 30 basis points as trading costs for all small stocks before 2001 and 40 basis points from 2001 on. For big stocks we use 15 basis points as trading costs for the whole time period. These values are adopted from Frazzini et al. (2012). In contrast to prior work (e.g. Lesmond et al., 2003)

which focuses on average trading costs for an average investor, Frazzini et al. (2012) are interested in the trading costs of a large arbitrageur. The authors use a proprietary dataset of live trading data from a large institutional money manager and estimate trading costs by comparing theoretical trades with trades actually executed.

We calculate the return r_t^{tc} for a long-short portfolio after trading costs as follows:

$$r_t^{tc} = r_t - tc^l \cdot to_t^l - tc^s \cdot to_t^s \quad (2)$$

where r_t is the portfolio return before trading costs in time t , tc^l (tc^s) denotes the trading costs of the long (short) portfolio, and to_t^l (to_t^s) is the portfolio turnover of the long (short) in time t . Trading costs of a portfolio are the result of portfolio turnover (stocks bought plus stocks sold in percentage of portfolio volume) times average trading costs. We apply this correction to all portfolios considered as well as to the portfolios used to construct the factors. This approach is only an approximation of the actual trading costs since we use fixed values and not an actual trading cost function like Frazzini et al. (2012). Given the constraints on available data, we consider this nevertheless as a useful exercise, especially in conjunction with our other tests, described in what follows.

In our second approach to assess the relevance of trading costs, we define what we call “critical trading costs”. The critical trading costs are the maximal trading costs that an investor can afford for a given spread portfolio so that the portfolio return is still (marginally) significantly positive for a certain significance level. We define the average critical trading costs tc_{crit} as:¹²

$$tc_{crit} = \frac{\frac{1}{T} \sum_{t=1}^T sp_t - t_{crit} \sqrt{\frac{1}{T} \sum_{t=1}^T (sp_t - \mu)^2}}{\frac{1}{T} \sum_{t=1}^T to_t} \quad (3)$$

Here, T is the number of periods, t_{crit} is the critical value of the assumed t-test (typically 1.64, 1.96 or

¹² Note that this is an approximate formula. The correct formula as well as the derivation of the approximate formula are discussed in appendix B. Using this approximation does not change the results qualitatively.

2.58 for 10%, 5%, or 1% significance levels), sp_t is the return of the spread portfolio with mean μ in time t , and to_t is the portfolio turnover in time t , as before. If the spread portfolio is long in small and short in big stocks, we still posit that the average trading costs of a big stock are 15 basis points. Since big stocks are more liquid than small stocks (e.g. De Moor and Sercu, 2013, Table 1), we believe this assumption is appropriate. When the long and short portfolios are averages of other portfolios, as it is the case with the SMB, and HML factors, the portfolio turnover is also an average of the respective portfolios.

Third, we calculate the monthly US dollar trading volume for each portfolio. This provides another way to assess whether the strategies under consideration actually work or may be infeasible due to illiquidity constraints. To calculate the US dollar trading volume we take the TRD variable ‘turnover by volume’ (VO), which represents the adjusted amount of shares traded per month and multiply it by the ‘price’ (P) variable converted into US dollars. Using the last price of each month and multiplying it by all traded shares of the whole month is an approximation. The exact approach would be to use the price for each transaction that was actually paid. However, we believe that this is not a crucial problem. A specific challenge can arise in the context of adjustments for corporate events. TRD calculates monthly figures from daily ones by summation, and hence sometimes mixes-up adjusted and non-adjusted figures, if these figures change within a given month. Therefore, we search for each country for the biggest two volume figures on the stock level and exclude these stocks completely if these high numbers seem inaccurate.^{13,14}

¹³ Unlike with return data these errors are not limited to small stocks. We also exclude rather big stocks such as Vodafone and Royal Dutch Shell in the UK. Since providing clean US dollar volume data is not the main focus of this paper we believe that this practice is appropriate and has little effect on the overall results.

¹⁴ To check if the US dollar volume data calculated this way are reasonable, we aggregated the TRD data for some of the countries under consideration and compared it with the aggregated US dollar volume data provided by the World Federation of Exchanges (WFE, <http://www.world-exchanges.org/statistics>). For most countries examined the US dollar volume data match up nicely. However, for Germany the figures are very different: The data from the WFE report a mean value of US\$148.67 billion for the period from 02/2002 to 12/2011 whereas the aggregated TRD data report a mean value of US\$4.60 billion. A possible remedy of this caveat would be to use the VC or VQ series of Datastream. However, the additional data would be only available from 2004 onwards, and there may be additional pitfalls.

3.2. *International size and momentum portfolios*

This section repeats the analysis of Section 2.1, but now accounting for trading costs using the three methods described in Section 3.1.

3.2.1 *Momentum conditional on size*

Table 6 shows the trading cost-adjusted raw returns of the momentum strategies by size groups. Primarily, we observe positive momentum premia for middle size stocks (columns (3) and (4)). Here, all premia except the Japanese one are positive and ten of them are significant (10% level). We also observe positive momentum premia for the other two size groups, but only few of them are significant (six for the small stocks and three for the big stocks). For the estimated risk adjusted momentum returns by size groups, shown in Table 7, all three groups show noteworthy estimated momentum premia. For big stocks we observe twelve positive momentum spreads (six of them are significant on the 5% level), for medium stocks we observe thirteen positive momentum spreads (ten of them are significant on the 5% level), and for small stocks we observe ten positive momentum spreads (six of them are significant on the 5% level). The differences between medium size stocks and big size stocks are similar as in section 2.1.1, as can be seen in columns (7) and (8) of the Tables 6 and 7.

[Table 6 here]

[Table 7 here]

Therefore, we omit Germany from the sample in this case.

We also report critical trading costs for the momentum premia. Table 8 shows the results. There are positive critical trading costs in every size group. In the biggest size group (columns (5) and (6)) we observe three countries, namely Australia, Canada, and Denmark, for which the critical trading costs are substantially higher than the trading costs imposed in the previous tables. For the middle size group we observe about ten countries with such substantial critical trading costs (columns (3) and (4)). For the small size group we observe six countries with substantial critical trading costs (columns (1) and (2)).¹⁵

[Table 8 here]

In Table 9 we show the average portfolio turnover of the nine portfolios sorted on size and momentum. The average monthly portfolio turnover for the loser and winner portfolios is between roughly 30% and 80%. Furthermore, in Table 10 we show the average monthly US dollar trading volume of the nine size and momentum portfolios. One can observe a large gap between the US dollar trading volume of the big stocks (columns 7-8) and the small stocks (columns 1-3). For example, for the US the big winner stocks show an average monthly trading volume of US\$788 billion, whereas the small winner stocks show an average monthly trading volume of only US\$537 million, about 1,500 times smaller. The volumes for the middle size stocks are in between, but still substantially smaller than for the big stocks.

To combine the previous results, we proceed in three steps. First, we have to determine what a reasonable portfolio or fund size for a large arbitrageur should be. Second, we determine the maximum fund size for our portfolio. Third, we set this into relation with the turnover volume of a

¹⁵ Note that the biggest of the critical trading costs shown here are much smaller than for the size strategies discussed in Schmidt et al. (2017). However, the middle size group has much smaller trading costs in reality, so that the assumed trading costs of 40 basis points for each trade is probably much smaller for most of the stocks in that group.

large arbitrageur.

First, Frazzini et al. (2012, Table 6) report implied fund sizes between US\$1.5 billion and US\$15 billion for the US. Therefore, we assume that a fund should have at least a size of US\$1.5 billion to be incepted.

As a second step, to compute the average maximum fund size, we utilize information on turnover and trading volume from Tables 9 and 10. We discuss the results for the US as an example. Consider small stocks first. The average monthly turnover of the small winner portfolio is 76% (Table 9, column (3), last row). If we assume that we invest in all available small size stocks in the US (with the proportions according to our value weights) we trade stocks with a volume of US\$527 million on average (Table 10, column (3), last row). Thus, we can conclude that the average maximum fund size (the 100%) is $\text{US\$527 million} / 0.76 \approx \text{US\$692 million}$. We can do the same exercise with the small loser portfolio and end up with an average maximum fund size of US\$1,304 million. Since the small winner portfolio has a smaller maximum fund size than the small loser portfolio, it limits the overall investment sum, since we want to allocate the same sum to both portfolios. Hence we only discuss the small winner portfolio here. Assuming that the hypothetical fund of US\$1.5 billion is equally divided between small and big stocks, we end up with a small size portfolio of about US\$750 million. This is more than the average maximum fund size computed above.

For the third step, to set this into relation with the turnover volume of a large arbitrageur, we are interested which fraction of the market can be traded under the assumptions made by Frazzini et al. (2012). Frazzini et al. (2012, Table 1, Panel B) report that for their overall trading data they move a fraction for about 1% of the daily trading volume. Assuming that this is the same on a monthly level, we conclude that the price impact of the small-size momentum strategy examined here is likely much higher.

For the medium-size stocks, implied by the medium loser portfolio, we calculate a maximum fund size of US\$15 billion (see column (4) of the last row in Tables 9 and 10). Given a fund size of US\$1.5 billion an investor will have to move about 10% of the market each month. If the induced price impact does not drive trading costs above 50 basis points on average (as implied by the critical trading costs in Table 8, columns (3) and (4) of the last row), then this strategy is able to provide positive returns on average.

For big size stocks, the maximum fund size is about US\$304 billion and therefore only a tiny fraction of this number has to be moved each month to reallocate the long-short portfolio.

[Table 9 here]

[Table 10 here]

We can also calculate the overall trading costs resulting from momentum conditional on size strategies. We compare the numbers from Section 2.1 with the numbers calculated in this section. As an example, consider the small momentum spread for the US (see Table 2, column (1), and Table 6, column (1)). The monthly overall trading costs are $0.68\% - 0.23\% = 0.45\%$, which is about 5.54% on a yearly basis. For comparison Mitchell and Pulvino (2001, p. 2136) report yearly overall trading costs of 5.71%. Since we apply a long-short strategy and Mitchell and Pulvino (2001) pursue a long-only strategy, our implied trading costs appear to be about half as large.¹⁶ This comparison shows that our approach of approximating trading costs yields roughly the same magnitude of trading costs as other studies. However, the stocks in the portfolio as well as the time period in Mitchell and

¹⁶ The portfolio turnover of the Mitchell and Pulvino (2001) strategy is similar as the turnover of our strategy. The stocks in the portfolio examined by Mitchell and Pulvino (2001) have an average duration of about two months, which would be a monthly portfolio turnover of 50%. Our strategy has a portfolio turnover of about 65% $(=(56\%+76\%)/2)$, see Table 9).

Pulvino (2001) is different than in our study.¹⁷ Therefore, we do not expect our results to be exactly the same.

3.2.2 *Summary*

In sum, momentum premia for small stocks are likely not realizable because of a small market depth.¹⁸ Momentum premia for medium size stocks are most likely feasible, though we cannot rule out that part of the returns are diminished due to price impact. Momentum premia for big stocks are present for some countries, but often smaller than for medium stocks.

4. Conclusion

Can investors profitably implement momentum strategies in international stock markets? Are there differences between different size groups? We document two sets of novel empirical results. First, we confirm, out-of-sample, the observation previously made by Hong et al. (2000) for US markets that the profitability of momentum strategies is higher for medium-sized stocks than for large stocks. This is the first international study that provides direct evidence of this empirical regularity. Future work may examine how momentum interacts with other differences among firms, such as information uncertainty. Also, there may be variation over time, depending, for example, on investor sentiment or market illiquidity. It is hard to study this time variation as one needs a sufficiently long time series to do so. Conceivably, for some countries, one can construct longer time series,

¹⁷ Mitchell and Pulvino (2001) study the “characteristics of risk and return in risk arbitrage”. Their sample ranges from 1963 to 1998.

¹⁸ In US data, Korajczyk and Sadka (2004) find that the profits of equal-weighted momentum strategies are significantly diminished after the consideration of trading costs, while the profits of value-weighted momentum strategies are affected by trading costs in a much smaller way. Therefore, their results also provide indirect evidence that small stocks are much more affected by trading costs than big stocks.

especially when paying the price of not including accounting data, thus focusing on risk-adjustment based solely on the CAPM.

Second, we incorporate trading costs in various ways into the analysis, therefore critically assessing how feasible it is to practically implement investment strategies based on size and momentum. When accounting for trading costs we find that all strategies involving small size stocks are probably not realizable because the US dollar trading volume of the small size stocks needed for implementation is too low and because actually trading these stocks with appropriate quantities would presumably increase stock prices and decrease the profitability of these strategies significantly. For the momentum strategies, which involve medium size and big stocks this problem is not that severe. Big stocks seem to provide a momentum premium for some countries. However, medium-size international stocks gain the biggest momentum premia even after trading costs.

Acknowledgements

The authors are grateful to Harald Lohre for useful comments and suggestions at the beginning of our project. Helpful research assistance of Katja Bächtold, Pascal Bücheler, Carolin Hecht, Marco Rudin and Veronika Sharonova is gratefully acknowledged. A special thanks goes to Tatjana-Xenia Puhan who helped us to expand the dataset to additional countries. Furthermore, we thank all researchers who contacted us regarding the data set. Their comments helped to improve the data set as well as the paper at hand.

This work was supported by the Swiss National Science Foundation [project 10CL14-120387]; the Commission for Technology and Innovation of the Swiss Confederation (CTI) [project 10603.2

PFES-ES]; the Swiss Finance Institute; the NCCR FINRISK [project B1: Corporate Finance, Market Structure and the Theory of the Firm]; and the Zurich Cantonal Bank (ZKB).

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Table 1: Position of this paper in the literature

	Sample	Factor model used for risk correction	Account of trading costs/market depth	Findings related to size-momentum (Winner minus Losers = WML)
Panel A: Studies on the interplay of size and momentum on US stock market data				
Jegadeesh and Titman (1993)	01/1965 - 12/1989, US	CAPM	50 BPs one-way transaction costs	Positive and significant WML spread returns for <u>all three size groups</u> , based on ten momentum groups
Hong, Lim, Stein (2000)	01/1980 - 12/1996, US	Market model on the stock level	State that results are robust to the inclusion of transaction cost proxies as controls	Positive and significant WML spread in <u>small size deciles, except for the smallest one</u> . Spread returns decline and become insignificant for higher market capitalization.
Jegadeesh and Titman (2001)	01/1965 - 12/1998, US	CAPM and Fama-French three factor model	No	Positive and significant WML spread returns for <u>both size groups</u> , based on ten momentum groups
Fama and French (2008)	07/1963 -12/2005, US	No factor model, but cross-sectional regressions (Fama-MacBeth)	No	Positive and significant WML spread returns for <u>three size groups</u> , based on four momentum groups
Avramov, Chordia, Jostova, Philipov (2013)	10/1985 - 12/2008, US	Returns are adjusted: Matching portfolio (25 portfolios, sorted by size and BE/ME)	No	Positive and significant WML spread returns for <u>all size groups</u> , based on five momentum groups
Israel and Moskowitz (2013)	01/1927 - 12/2011, US	CAPM	Examine whether changes in aggregate trading costs affect premia	Positive WML spread returns for <u>all ten size groups</u> , based on five momentum groups (not explicitly reported, hence no t-statistic is available)

Panel B: Studies on the interplay of size and momentum on international stock markets

Fama and French (2012)	11/1990 - 03/2011, Regional data: North America, Europe, Japan, and Asia Pacific (excluding Japan)	Carhart four-factor model	No	Positive WML spread returns for the <u>two smallest of five size groups</u> , based on five momentum groups, for all regions examined, except Japan (a spread return is not explicitly reported, hence no t-statistic is available)
Cakici and Tan (2014)	01/1991 - 03/2012, North America, Europe, Japan, Australia, Singapore, Hong Kong, New Zealand	None	No	Positive and significant WML spread returns for the <u>smaller of two size groups</u> , based on three momentum groups
This Paper	07/1991 - 02/2012, North America, Europe, Japan, Australia, Singapore, Hong Kong, Turkey	Fama-French three-factor model	1. Use fixed trading costs, 2. calculate critical trading costs, 3. use a measure for market depth	Positive and significant WML spread returns for <u>medium</u> size groups (from three size groups), based on three momentum groups for most countries examined

Table 2: Momentum returns (relative to size) – bivariate sorts

	W-L small		W-L med		W-L big		W-L med – W-L big	
	Mean	t	Mean	t	Mean	t	Mean	t
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Australia	0.11	0.29	1.80	5.96	1.34	3.63	0.46	1.36
Canada	1.16	2.55	2.34	6.60	1.58	3.03	0.76	1.90
Denmark	1.19	3.51	1.15	3.63	1.28	3.12	0.07	0.16
France	1.20	4.08	1.25	3.97	0.50	1.45	0.74	2.47
Germany	0.84	2.51	1.79	5.46	0.71	1.77	0.99	2.69
Hong Kong	1.01	1.45	1.23	2.28	0.51	0.79	0.72	1.17
Italy	1.25	2.70	0.90	2.97	0.35	0.91	0.55	1.47
Japan	-0.02	-0.09	-0.01	-0.05	0.09	0.25	-0.11	-0.38
Netherlands	2.16	5.88	1.16	2.95	0.35	0.75	0.81	1.77
Norway	0.70	1.32	0.98	2.17	1.16	2.43	-0.18	-0.36
Singapore	-0.19	-0.40	0.55	1.34	-0.05	-0.10	0.60	1.50
Switzerland	1.11	4.08	1.15	3.79	0.43	1.13	0.73	2.23
UK	1.46	5.17	1.67	5.62	0.92	2.05	0.75	2.12
US	0.68	2.24	1.19	3.12	0.68	1.54	0.52	2.06

Note: We report average raw returns of three momentum spreads long in winners and short in losers (shorthand W-L) of a double sort on size and momentum for small (W-L small), medium (W-L med), big (W-L big) stocks, and the difference of the medium and big momentum spread (W-L med – W-L big), as well as the corresponding t-statistics. t-statistics refer to the null that the means are equal to zero. Momentum is calculated using the 12 previous months, skipping the most recent one. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Singapore, Switzerland, UK, US; 07/2000-02/2012: Hong Kong; 03/1993-02/2012: Japan; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 3: Risk-adjusted momentum returns (relative to size) – bivariate sorts

	W-L small		W-L med		W-L big		W-L med – W-L big	
	Alpha	t	Alpha	t	Alpha	t	Alpha	t
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Australia	0.13	0.34	1.87	5.98	1.32	3.52	0.55	1.65
Canada	1.12	2.58	2.54	6.86	1.74	3.04	0.81	1.95
Denmark	1.26	3.75	1.51	4.63	1.43	3.70	0.08	0.22
France	1.35	4.67	1.43	4.96	0.78	2.61	0.65	2.53
Germany	0.81	2.56	1.81	6.08	1.15	2.70	0.66	1.83
Hong Kong	1.33	1.91	1.38	2.59	0.64	0.90	0.74	1.26
Italy	1.28	2.89	0.95	3.14	0.43	1.10	0.52	1.56
Japan	-0.07	-0.27	-0.02	-0.06	0.19	0.48	-0.21	-0.68
Netherlands	2.23	6.22	1.31	2.90	0.56	1.37	0.75	1.75
Norway	1.03	2.06	1.15	2.40	1.23	2.66	-0.08	-0.14
Singapore	-0.07	-0.17	0.75	1.79	0.29	0.68	0.45	1.11
Switzerland	1.31	4.87	1.29	3.95	0.49	1.24	0.80	2.41
UK	1.71	6.26	2.01	6.34	1.44	3.58	0.57	1.52
US	0.76	2.31	1.57	4.80	1.14	3.20	0.43	1.81

Note: We report estimated regression slopes (or three-factor alphas) of the following regression: $(r_t^{w,g} - r_t^{l,g}) = \alpha + \beta_M * (r_{M,t} - r_{f,t}) + \beta_{SMB} * SMB_t + \beta_{HML} * HML_t + \varepsilon_t$

where $r_t^{w,g}$ ($r_t^{l,g}$) is the return of the winner (loser) tercile of size group g in time t , $r_{M,t}$ is the return of the market portfolio in time t , $r_{f,t}$ is the riskfree rate proxy in time t , SMB_t and HML_t are the factors as described in Appendix A, β_M , β_{SMB} , and β_{HML} are the corresponding factor loadings, α is the four factor alpha, and ε_t is an error term. g =small for columns (1) and (2), g =medium (med) for columns (3) and (4), and g =big for columns (5) and (6). In addition, we report the difference of the medium and big momentum spread (W-L med – W-L big) in columns (7) and (8). We report heteroscedasticity and autocorrelation robust t-statistics according to Newey and West (1987) with three lags for the correction of autocorrelation. t-statistics refer to the null that the alphas are equal to zero. Momentum is calculated using the 12 previous months, skipping the most recent one. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Singapore, Switzerland, UK, US; 07/2000-02/2012: Hong Kong; 03/1993-02/2012: Japan; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 4: Returns of nine portfolios sorted on size and momentum

	Losers	Small Med	Winners	Losers	Med Med	Winners	Losers	Big Med	Winners
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Australia	2.28 (4.41)	2.43 (5.03)	2.39 (4.46)	-0.66 (-1.53)	0.22 (0.69)	1.14 (2.70)	-0.63 (-1.41)	0.19 (0.67)	0.71 (2.27)
Canada	2.42 (4.29)	2.46 (5.43)	3.59 (6.07)	-0.35 (-0.70)	0.82 (2.55)	1.99 (4.73)	-0.36 (-0.69)	0.43 (1.33)	1.22 (3.08)
Denmark	-0.02 (-0.06)	0.37 (1.49)	1.17 (3.66)	-0.51 (-1.24)	0.19 (0.68)	0.84 (2.54)	-0.25 (-0.53)	0.37 (0.97)	1.03 (3.16)
France	0.49 (1.37)	0.61 (2.75)	1.69 (5.98)	0.01 (0.03)	0.36 (1.48)	1.26 (4.26)	0.10 (0.22)	0.58 (1.82)	0.61 (1.87)
Germany	-0.20 (-0.53)	0.16 (0.67)	0.64 (2.60)	-1.08 (-2.64)	-0.07 (-0.33)	0.71 (3.21)	-0.11 (-0.21)	0.27 (0.76)	0.70 (2.17)
Hong Kong	2.89 (3.18)	2.75 (3.67)	3.89 (4.25)	0.26 (0.30)	0.78 (1.12)	1.49 (2.15)	0.08 (0.10)	0.11 (1.82)	0.59 (1.05)
Italy	-0.33 (-0.67)	0.08 (0.22)	0.92 (1.99)	-0.38 (-0.75)	0.07 (0.18)	0.53 (1.32)	0.04 (0.07)	0.17 (0.41)	0.39 (0.89)
Japan	0.44 (1.01)	0.38 (1.09)	0.41 (1.11)	0.05 (0.10)	0.06 (0.17)	0.03 (0.09)	-0.13 (-0.27)	-0.30 (-0.81)	-0.04 (-0.11)
Netherlands	-0.56 (-1.39)	0.65 (2.08)	1.60 (4.09)	0.09 (0.20)	0.85 (2.44)	1.25 (3.22)	0.35 (0.65)	0.54 (1.49)	0.70 (1.85)
Norway	0.75 (1.38)	0.75 (1.93)	1.45 (3.22)	0.36 (0.61)	1.07 (2.85)	1.34 (2.82)	-0.18 (-0.29)	0.75 (1.67)	0.98 (1.96)
Singapore	1.50 (2.05)	1.26 (1.97)	1.32 (2.21)	0.32 (0.45)	0.74 (1.29)	0.87 (1.63)	0.73 (1.13)	0.83 (1.61)	0.68 (1.61)
Switzerland	0.26	0.63	1.37	0.06	0.62	1.21	0.34	0.61	0.77

	(0.80)	(2.68)	(5.27)	(0.14)	(2.33)	(3.95)	(0.77)	(1.98)	(2.36)
UK	0.12	0.40	1.58	-0.50	0.07	1.17	-0.22	0.36	0.71
	(0.30)	(1.48)	(5.07)	(-1.11)	(0.23)	(3.59)	(-0.41)	(1.15)	(2.43)
US	1.71	1.63	2.39	-0.02	0.66	1.18	0.12	0.45	0.79
	(3.12)	(5.16)	(5.84)	(-0.03)	(2.10)	(3.12)	(0.21)	(1.43)	(2.69)

Note: We report average raw excess returns of nine portfolios sorted on size and momentum. Columns (1)-(3) show the one third smallest stocks, columns (4)-(6) show the middle third of stocks based on size, and columns (7)-(9) show the one third of biggest stocks. Within these three groups each first column shows the one third of stocks with the lowest momentum (losers), the second column shows the middle third of stocks based on momentum, and the last column shows the one third with the highest momentum (winners). Momentum is calculated using the 12 previous months, skipping the most recent one. The numbers in parentheses represent t-statistics. t-statistics refer to the null that the average excess returns are equal to zero. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Singapore, Switzerland, UK, US; 07/2000-02/2012: Hong Kong; 03/1993-02/2012: Japan; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 5: Risk-adjusted returns of nine portfolios sorted on size and momentum

	Losers	Small Med	Winners	Losers	Med Med	Winners	Losers	Big Med	Winners
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Australia	1.48 (3.61)	1.67 (3.93)	1.62 (3.39)	-1.45 (-6.80)	-0.43 (-2.82)	0.42 (1.52)	-1.12 (-3.87)	-0.24 (-1.74)	0.20 (1.15)
Canada	1.46 (3.23)	1.64 (4.14)	2.58 (5.89)	-1.43 (-5.15)	0.04 (0.25)	1.12 (4.81)	-1.23 (-2.70)	-0.20 (-1.16)	0.51 (2.89)
Denmark	-0.32 (-1.24)	0.19 (1.01)	0.94 (3.63)	-0.91 (-4.37)	-0.06 (-0.43)	0.60 (2.67)	-0.82 (-2.93)	-0.13 (-0.74)	0.60 (3.47)
France	-0.07 (-0.29)	0.21 (1.52)	1.28 (6.02)	-0.64 (-2.85)	-0.15 (-1.43)	0.79 (5.01)	-0.55 (-2.57)	0.13 (1.02)	0.23 (1.95)
Germany	-0.24 (-1.08)	-0.03 (-0.16)	0.57 (2.61)	-1.17 (-5.29)	-0.24 (-1.97)	0.64 (4.25)	-0.69 (-2.30)	-0.18 (-1.30)	0.47 (2.74)
Hong Kong	1.21 (2.47)	1.38 (2.92)	2.54 (3.57)	-1.20 (-2.60)	-0.56 (-1.56)	0.18 (0.43)	-0.75 (-1.43)	0.50 (1.83)	-0.12 (-0.38)
Italy	-0.51 (-1.91)	-0.08 (-0.41)	0.76 (2.10)	-0.61 (-2.74)	-0.13 (-0.80)	0.35 (1.80)	-0.23 (-0.93)	-0.05 (-0.27)	0.19 (1.07)
Japan	0.52 (2.53)	0.37 (3.01)	0.44 (2.35)	0.08 (0.36)	0.02 (0.19)	0.06 (0.40)	-0.11 (-0.43)	-0.27 (-2.27)	0.08 (0.50)
Netherlands	-1.07 (-4.01)	0.22 (0.87)	1.16 (3.66)	-0.57 (-1.58)	0.26 (1.41)	0.74 (2.75)	-0.39 (-1.40)	0.05 (0.31)	0.17 (0.88)
Norway	-0.23 (-0.71)	0.04 (0.16)	0.79 (2.16)	-0.74 (-2.29)	0.29 (1.15)	0.41 (1.48)	-1.10 (-3.21)	-0.04 (-0.21)	0.86 (1.88)
Singapore	0.86 (2.21)	0.62 (2.02)	0.78 (2.28)	-0.44 (-1.63)	0.02 (0.14)	0.31 (1.38)	-0.16 (-0.47)	-0.06 (-0.24)	0.13 (0.78)
Switzerland	-0.53	0.10	0.78	-0.92	-0.07	0.38	-0.45	-0.03	0.05

	(-2.24)	(0.55)	(4.68)	(-3.58)	(-0.44)	(2.16)	(-1.70)	(-0.20)	(0.28)
UK	-0.46	-0.02	1.25	-1.28	-0.47	0.73	-1.05	-0.12	0.39
	(-2.03)	(-0.13)	(5.81)	(-5.89)	(-3.59)	(4.87)	(-3.29)	(-1.17)	(3.05)
US	1.10	1.17	1.86	-1.08	-0.11	0.49	-0.87	-0.18	0.27
	(2.45)	(4.33)	(5.92)	(-4.03)	(-0.86)	(2.55)	(-3.09)	(-1.92)	(2.73)

Note: We report estimated regression slopes (or three-factor alphas) of the following regression: $(r_t^{g,m} - r_{f,t}) = \alpha + \beta_M * (r_{M,t} - r_{f,t}) + \beta_{SMB} * SMB_t + \beta_{HML} * HML_t + \varepsilon_t$ where $r_t^{g,m}$ is the return of a portfolio where the stock are both contained in the size tercile g and the momentum tercile m in time t , $r_{M,t}$ is the return of the market portfolio in time t , $r_{f,t}$ is the riskfree rate proxy in time t , SMB_t and HML_t are the factors as described in Appendix , β_M , β_{SMB} , and β_{HML} are the corresponding factor loadings, α is the four-factor alpha, and ε_t is an error term. g =small (columns (1)-(3)), g =medium (med) (columns (4)-(6), and g =big (columns (7)-(9)). m =losers (columns (1), (4) and (7)), m =medium (med) (columns (2), (5) and (8)), and m =winners (columns (3),(6) and (9)). Momentum is calculated using the 12 previous months, skipping the most recent one. The numbers in parentheses represent heteroscedasticity and autocorrelation robust t-statistics according to Newey and West (1987) with three lags for the correction of autocorrelation. t-statistics refer to the null that the alphas are equal to zero. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Singapore, Switzerland, UK, US; 07/2000-02/2012: Hong Kong; 03/1993-02/2012: Japan; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 6: Momentum returns (relative to size), trading cost-adjusted– bivariate sorts

	W-L small		W-L med		W-L big		W-L med – W-L big	
	Mean	t	Mean	t	Mean	t	Mean	t
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Australia	-0.35	-0.92	1.48	4.90	1.04	2.81	0.44	1.33
Canada	0.74	1.62	2.03	5.72	1.32	2.53	0.71	1.76
Denmark	0.85	2.51	1.04	2.80	1.04	2.54	0.00	-0.01
France	0.84	2.85	0.94	2.98	0.23	0.66	0.70	2.35
Germany	0.48	1.43	1.49	4.55	0.54	1.20	0.95	2.58
Hong Kong	0.77	1.04	0.92	1.60	0.38	0.57	0.54	0.90
Italy	0.90	1.94	0.58	1.92	0.06	0.17	0.52	1.39
Japan	-0.43	-1.75	-0.36	-1.39	-0.16	-0.44	-0.20	-0.72
Netherlands	1.83	5.01	0.86	2.18	0.04	0.08	0.82	1.79
Norway	0.31	0.58	0.62	1.37	0.79	1.65	-0.17	-0.34
Singapore	-0.61	-1.32	0.20	0.49	-0.44	-0.87	0.64	1.61
Switzerland	0.77	2.83	0.87	2.85	0.11	0.29	0.76	2.32
UK	1.11	3.93	1.38	4.63	0.63	1.40	0.75	2.11
US	0.23	0.75	0.87	2.28	0.41	0.94	0.46	1.84

Note: We report average raw returns of three momentum spreads long in winners and short in losers (shorthand W-L) of a double sort on size and momentum for small (W-L small), medium (W-L med), big (W-L big) stocks, and the difference of the medium and big momentum spread (W-L med – W-L big), as well as the corresponding t-statistics. Momentum is calculated using the 12 previous months, skipping the most recent one. t-statistics refer to the null that the means are equal to zero. We assume that trading costs for small stocks are 30 basis points before 2001 and 40 basis points from 2001 on. For big stocks we assume trading costs of 15 basis points over the whole time period. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Singapore, Switzerland, UK, US; 07/2001-02/2012: Hong Kong; 03/1993-02/2012: Japan; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 7: Risk-adjusted momentum returns (relative to size), trading cost-adjusted– bivariate sorts

	W-L small		W-L med		W-L big		W-L med – W-L big	
	Alpha	t	Alpha	t	Alpha	t	Alpha	t
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Australia	-0.34	-0.87	1.54	4.94	1.01	2.73	0.53	1.60
Canada	0.70	1.60	2.22	5.95	1.47	2.56	0.76	1.83
Denmark	0.92	2.71	1.19	3.65	1.16	2.99	0.03	0.07
France	0.98	3.38	1.11	3.86	0.49	1.67	0.62	2.42
Germany	0.44	1.38	1.50	5.07	0.86	2.03	0.63	1.76
Hong Kong	0.97	1.28	1.12	1.93	0.63	0.88	0.49	0.85
Italy	0.93	2.12	0.64	2.11	0.13	0.33	0.51	1.53
Japan	-0.48	-1.78	-0.37	-1.29	-0.08	-1.20	-0.29	-0.96
Netherlands	1.91	5.32	1.00	2.20	0.25	0.61	0.75	1.74
Norway	0.63	1.27	0.78	1.63	0.86	1.86	-0.08	-0.15
Singapore	-0.52	-1.24	0.38	0.90	-0.11	-0.25	0.49	1.19
Switzerland	0.98	3.60	1.01	3.08	0.19	0.49	0.82	2.44
UK	1.35	4.89	1.70	5.32	1.12	2.78	0.58	1.54
US	0.31	0.93	1.28	3.79	0.87	2.44	0.38	1.61

Note: We report estimated regression slopes (or three-factor alphas) of the following regression: $(r_t^{w,g} - r_t^{l,g}) = \alpha + \beta_M * (r_{M,t} - r_{f,t}) + \beta_{SMB} * SMB_t + \beta_{HML} * HML_t + \varepsilon_t$ where $r_t^{w,g}$ ($r_t^{l,g}$) is the return of the winner (loser) tercile of size group g in time t , $r_{M,t}$ is the return of the market portfolio in time t , $r_{f,t}$ is the riskfree rate proxy in time t , SMB_t and HML_t are the factors as described in Appendix A, β_M , β_{SMB} , and β_{HML} are the corresponding factor loadings, α is the four factor alpha, and ε_t is an error term. g =small for columns (1) and (2), g =medium (med) for columns (3) and (4), and g =big for columns (5) and (6). In addition, we report the difference of the medium and big momentum spread (W-L med – W-L big) in columns (7) and (8). We report heteroscedasticity and autocorrelation robust t-statistics according to Newey and West (1987) with three lags for the correction of autocorrelation. t-statistics refer to the null that the means are equal to zero. Momentum is calculated using the 12 previous months, skipping the most recent one. We assume that trading costs for small stocks are 30 basis points before 2001 and 40 basis points from 2001 on. For big stocks we assume trading costs of 15 basis points over the whole time period. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Singapore, Switzerland, UK, US; 07/2001-02/2012: Hong Kong; 03/1993-02/2012: Japan; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 8: Critical trading costs of momentum returns (relative to size) – bivariate sorts

	W-L small		W-L med		W-L big	
	10%	5%	10%	5%	10%	5%
	(1)	(2)	(3)	(4)	(5)	(6)
Australia	neg	neg	131.97	122.36	65.96	55.54
Canada	33.68	21.99	182.26	170.69	73.62	56.97
Denmark	65.41	54.39	79.95	67.31	68.59	54.02
France	68.46	59.65	77.29	66.83	neg	neg
Germany	28.09	17.94	138.10	126.73	6.00	neg
Hong Kong	5.47	neg	34.47	17.83	neg	neg
Italy	46.78	32.88	41.22	31.49	neg	neg
Japan	neg	neg	neg	neg	neg	neg
Netherlands	168.66	156.16	55.65	42.29	neg	neg
Norway	neg	neg	21.51	8.65	29.75	17.94
Singapore	neg	neg	neg	neg	neg	neg
Switzerland	68.33	59.53	76.30	65.12	neg	neg
UK	98.99	90.18	131.12	120.75	17.01	3.96
US	13.75	6.56	57.01	44.86	neg	neg

Note: We calculate critical trading costs as follows:

$$tc_{crit} = \frac{\frac{1}{T} \sum_{t=1}^T sp_t - t_{crit} \sqrt{\frac{1}{T} \sum_{t=1}^T (sp_t - \mu)^2}}{\frac{1}{T} \sum_{t=1}^T to_t}$$

with T as the number of periods, t_{crit} as the critical value of the assumed t-test, sp_t as the return of the spread portfolio with mean μ , and to_t as the portfolio turnover. We report the critical trading costs for three momentum spreads long in winners and short in losers (shorthand W-L) of a double sort on size and momentum for small (W-L small), medium (W-L med), and big (W-L big) stocks. The critical trading costs are denoted in basis points. We report critical trading costs for two significance levels: 10% (columns (1), (3) and (5)) and 5% (columns (2), (4) and (6)). Since we report only positive critical trading costs we denote negative values simply with 'neg'. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Japan, Singapore, Switzerland, UK, US; 07/2001-02/2012: Hong Kong; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 9: Turnover of nine portfolios sorted on size and momentum

	Small			Med			Big		
	Losers	Med	Winners	Losers	Med	Winners	Losers	Med	Winners
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Australia	55.90	102.74	75.99	51.98	76.58	46.91	70.26	39.76	41.38
Canada	53.29	99.95	69.57	50.33	72.54	46.13	60.64	38.91	37.71
Denmark	41.91	66.22	55.05	46.37	65.76	45.99	56.73	73.32	31.89
France	44.19	85.68	60.33	48.74	68.96	45.83	59.52	59.98	43.08
Germany	45.86	77.82	58.41	46.89	69.89	43.88	61.34	55.55	36.09
Hong Kong	54.21	85.64	76.63	55.82	78.00	52.69	60.34	42.29	41.54
Italy	46.98	79.30	57.75	51.75	76.59	46.41	68.52	61.19	37.78
Japan	52.42	84.12	66.62	52.78	80.20	52.34	58.34	75.16	39.74
Netherlands	42.24	79.48	49.93	47.13	68.14	45.41	60.65	69.22	51.78
Norway	46.95	85.84	62.28	56.75	79.87	53.89	79.17	67.43	48.19
Singapore	50.73	88.71	70.56	53.51	77.31	52.21	91.69	76.01	45.54
Switzerland	41.87	76.58	55.10	44.47	72.77	41.13	66.68	74.54	46.43
UK	44.32	79.30	56.55	47.37	69.14	42.81	63.82	50.25	44.49
US	56.32	94.90	76.07	51.20	68.88	48.02	64.83	42.10	37.29

Note: We report the average turnover of nine portfolios sorted on size and momentum. We calculate turnover based on a value weighted scheme, according to the corresponding portfolios. Turnover is reported in percent and on a monthly basis. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, France, Germany, Italy, Japan, Singapore, Switzerland, UK, US; 07/2001-02/2012: Hong Kong; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

Table 10: Trading volume of nine portfolios sorted on size and momentum

	Losers (1)	Small Med (2)	Winners (3)	Losers (4)	Med Med (5)	Winners (6)	Losers (7)	Big Med (8)	Winners (9)
Australia	28.41	15.19	22.93	138.48	94.49	210.44	3,670.25	21,213.37	13,175.97
Canada	63.37	30.19	50.41	358.49	323.41	571.34	6,685.70	20,364.22	18,836.29
Denmark	7.83	5.21	6.83	48.57	31.83	41.30	1,235.66	1,643.12	2,462.26
France	29.57	9.93	28.10	147.35	115.76	161.36	16,532.48	35,285.23	25,494.07
Hong Kong	114.49	52.75	78.33	334.32	286.25	549.56	6,011.45	14,624.70	12,548.78
Italy	128.36	76.96	109.40	337.16	288.11	433.86	11,788.77	25,278.27	20,156.41
Japan	472.86	269.74	487.42	2,313.88	1,836.80	3,664.68	48,082.50	54,841.40	85,428.32
Netherlands	33.11	29.25	33.51	380.93	289.26	343.76	10,699.57	17,071.01	15,712.39
Norway	31.14	15.38	24.51	156.52	102.25	180.18	1,856.13	4,370.07	5,281.77
Singapore	35.30	26.76	49.37	128.29	130.57	262.20	1,326.70	3,060.99	3,247.56
Switzerland	33.15	21.24	25.82	260.25	166.87	254.52	12,994.38	15,996.27	11,847.22
UK	94.10	29.74	39.35	508.56	356.41	579.70	24,078.50	75,610.70	58,974.01
US	734.59	228.59	526.67	7,697.13	4,677.37	11,571.33	193,840.94	682,671.23	787,836.47

Note: We report the average trading volume of the nine portfolios sorted on size and momentum. We calculate trading volume by multiplying the TRD variable turnover by volume (VO) with the adjusted price (P) on a monthly basis. Trading volume is reported in million US\$ on a monthly basis. We use the following time periods: 07/1991-02/2012: Australia, Canada, Denmark, Japan, Singapore, Switzerland, UK, US; 01/1992-02/2012: France; 05/1994-02/2012: Italy; 07/2001-02/2012: Hong Kong; 07/1991-06/2011: Netherlands; 07/1993-02/2012: Norway.

A. Supplementary Appendix

Please note: The following supplementary appendices are not meant for publication in print. They are included for the benefit of the Referees.

Appendix A. Construction of common risk factors

This appendix describes the construction of common risk factors as proposed by Fama and French (1993).

Fama and French (1993) introduced common risk factors based on individual stock characteristics. To obtain market-wide factors from individual firm characteristics, Fama and French (1993) sorted stocks on these characteristics and used the difference in portfolio returns between high rated and low rated stocks according to these characteristics. In particular, they proposed one factor based on the difference in portfolio returns between stocks with a small market capitalization and stocks with a big market capitalization (small-minus-big – SMB) and one factor based on the difference between stocks with a high book-to-market equity ratio and a low book-to-market equity ratio (high-minus-low – HML) in addition to the market factor. This empirical model has become standard in the empirical asset pricing literature. We follow this method to construct the factors SMB and HML.

Our TRD and TRW dataset of monthly observations begins in December 1985 and ends in February 2012.¹⁹ The return calculation is based on closing prices of the last trading day of each month. If a stock is not traded on the last trading day, the last valid trading price is used. The TRD total return indices which we use for return calculation include dividends and account for stock splits.

¹⁹ Note that we therefore begin with the portfolio formation in June 1986 and with the calculation of return series in July 1986. This applies only to the US dataset. For other countries a different time span is used.

Book equity is TRW ‘common equity’ (WC03501) in our dataset. For sorts utilizing book equity we use only stocks with available book equity which is greater than zero. Size is either the TRD ‘market value’ (MV) or the product of the TRD ‘unadjusted price’ (UP) with the TRD ‘number of shares’ (NOSH). BE/ME for the sorting month June is calculated as book equity of the previous fiscal year divided by size of the preceding December. For the construction of the SMB and HML factors, we sort all stocks each June, beginning in 1986. To be included in the June sort of year τ a stock must have an available and positive book value and size available in December of the previous year $\tau-1$. Furthermore, to calculate value weighted returns, a stock needs to have available size from the preceding month, a valid return, an available and positive book value, as well as ‘price’ available and ‘number of shares’ different from zero.²⁰

In order to construct the SMB and HML factors, all remaining stocks are sorted each December into three BE/ME groups (breakpoints are discussed in Schmidt et al. (2017); for the size sort we use the 0.8 quantile as breakpoint, for the BE/ME sort we use equal breakpoints). Furthermore, we sort these stocks each June into two size groups. From the intersection of the two size groups, small (S) and big (B), and the three BE/ME groups, low (L), medium (M) and high (H), we form six portfolios, which are held for one year.²¹ Table A.1 illustrates the sorting procedure.

[Table A.1 here]

From the monthly value weighted returns of these six portfolios we construct the factors SMB and HML for month t as follows:

²⁰ For sorts which do not utilize book equity, we do not require that book value is available and positive. However, certainly the characteristic on which the sorting is done has to be available in addition to the other mentioned requirements.

²¹ When a stock is no longer available in our dataset we invest the share of this stock into the other stocks in the respective portfolio group according to the employed weighting scheme.

$$SMB_t = \frac{r_t^{S/L} + r_t^{S/M} + r_t^{S/H}}{3} - \frac{r_t^{B/L} + r_t^{B/M} + r_t^{B/H}}{3}, \quad (A.1)$$

$$HML_t = \frac{r_t^{S/H} + r_t^{B/H}}{2} - \frac{r_t^{S/L} + r_t^{B/L}}{2}. \quad (A.2)$$

Where $r_t^{X/Y}$ denotes the returns of a portfolio of stocks belonging to size class X (either S or B) and BE/ME class Y (either H, M or L) in month t based on the portfolio formation in last June.

Appendix B. Critical trading costs – derivation of the formulas

This appendix describes the formulas used for the critical trading costs of section 3.1 in detail.

We calculate the critical trading costs for a portfolio return denoted r_t , for our applications which is either the return of a long-short portfolio or the return of a long portfolio in excess of the riskfree rate. Thereby we employ the portfolio turnover to_t to approximate the trading costs by assuming constant trading costs tc for each trade. The effective trading costs are therefore the product of the trading costs and the portfolio turnover $tc \cdot to_t$. For a long portfolio in excess of the riskfree rate the portfolio turnover consists only of the turnover of the long portfolio. For a long-short portfolio the portfolio turnover is the sum of the portfolio turnover of the long and the short portfolio ($to_t = to_t^l + to_t^s$). When we go long in a portfolio of small stocks and short in a portfolio of big stocks, we assume a certain value for the trading costs of the big portfolio (which is 15 basis points), and the critical trading costs apply only to the small portfolio. In this case, we subtract the trading costs of the big portfolio from the overall portfolio return and r_t is then the return of the long portfolio minus the return of the short portfolio minus the effective trading costs of the short portfolio. The portfolio turnover is in this case the portfolio turnover of the long (small) portfolio.

Next, we define a few shorthand notations for the ease of exposition later. First we define the (empirical) means of the return and the turnover of the (spread) portfolio:

$$\mu_{sp} = \frac{1}{T} \sum_{t=1}^T r_t \quad \mu_{to} = \frac{1}{T} \sum_{t=1}^T to_t \quad (B.1)$$

And second the (empirical) variances of the return and turnover of the (spread) portfolio as well as the (empirical) covariance between the return and turnover:

$$\sigma_{sp}^2 = \frac{1}{T} \sum_{t=1}^T (r_t - \mu_{sp})^2 \quad \sigma_{to}^2 = \frac{1}{T} \sum_{t=1}^T (to_t - \mu_{to})^2 \quad (B.2)$$

$$\sigma_{sp,to} = \frac{1}{T} \sum_{t=1}^T (r_t - \mu_{sp}) \cdot (to_t - \mu_{to}) \quad (B.3)$$

Next we look at the definition of the critical value. In our case this is the following:

$$t_{crit} = \frac{\frac{1}{T} \sum_{t=1}^T (r_t - tc \cdot to_t)}{\sqrt{\frac{1}{T} \sum_{t=1}^T (r_t - tc \cdot to_t - \mu_{sp} - \mu_{to})^2}} \quad (B.4)$$

In the following we assume that the critical value t_{crit} is given and that the trading costs tc are a function of the critical value. Our aim is therefore to determine the value of the trading costs tc for a given critical value t_{crit} .

By slightly rearranging this formula and inserting the expressions above this expression simplifies to

$$t_{crit} = \frac{\mu_{sp} - tc \cdot \mu_{to}}{\sqrt{\sigma_{sp}^2 - 2 \cdot tc \cdot \sigma_{sp,to} + tc^2 \cdot \sigma_{to}^2}} \quad (B.5)$$

If we multiply both sides by the denominator of the right hand side, divide by t_{crit} and take the square this yields

$$\sigma_{sp}^2 - 2 \cdot tc \cdot \sigma_{sp,to} + tc^2 \cdot \sigma_{to}^2 = \frac{\mu_{sp}^2 - 2 \cdot tc \cdot \mu_{sp} \cdot \mu_{to} + tc^2 \cdot \mu_{to}^2}{t_{crit}^2} \quad (B.6)$$

We then rearrange this expression to obtain a quadratic equation:

$$tc^2 \cdot \left(\sigma_{to}^2 - \left(\frac{\mu_{to}}{t_{crit}} \right)^2 \right) - 2 \cdot tc \left(\sigma_{sp,to} - \frac{\mu_{sp} \cdot \mu_{to}}{t_{crit}^2} \right) + \sigma_{sp}^2 - \left(\frac{\mu_{sp}}{t_{crit}} \right)^2 = 0. \quad (B.7)$$

Hence, we obtain the following solutions for tc :

$$tc_{1,2} = \frac{\left(\sigma_{sp,to} - \frac{\mu_{sp}\mu_{to}}{t_{crit}^2}\right) \pm \sqrt{\left(\sigma_{sp,to} - \frac{\mu_{sp}\mu_{to}}{t_{crit}^2}\right)^2 - \left(\sigma_{to}^2 - \left(\frac{\mu_{to}}{t_{crit}}\right)^2\right)\left(\sigma_{sp}^2 - \left(\frac{\mu_{sp}}{t_{crit}}\right)^2\right)}}{\sigma_{to}^2 - \left(\frac{\mu_{to}}{t_{crit}}\right)^2}$$

(B.8)

We can obtain an approximate solution if we set $\sigma_{sp,to}=0$ and $\sigma_{to}^2=0$. With a little rearrangement, we then obtain the following approximate formula for the critical trading costs:

$$tc_{1,2} = \frac{\mu_{sp} \pm t_{crit} \cdot \sigma_{sp}}{\mu_{to}} \quad (B.9)$$

This formula is very similar to the formula of a confidence interval; the only difference is that we divide by the mean portfolio turnover μ_{to} . Since we aim to find out how much trading costs we can afford and still gain a positive mean return it makes sense to subtract the $t_{crit} \cdot \sigma_{sp}$ term from μ_{sp} . Note that this is the same as adding the term on the right hand side of equation (B.8) since the term on the left in the nominator and the term in the denominator are both negative. We then obtain the following formula:

$$tc = \frac{\mu_{sp} - t_{crit} \cdot \sigma_{sp}}{\mu_{to}} \quad (B.10)$$

Note that this approximate formula has several advantages over the formula in equation (B.8). First, it is much simpler than the formula in equation (B.8) and it is always obvious whether to subtract or add the square root term (Note that in our applications we always subtract the square root term. However, in general it is unclear whether this is necessarily the case in all circumstances). Second, with this formula we always obtain a real solution. The formula in equation (B.8) can also yield complex solutions, which is a discouraging result (In our applications we only observe one case where this happens. It seems that these cases occur when the critical trading costs are supposed to be negative anyway, so that a real solution is of minor importance in those cases). Third, the approximate formula has a rather intuitive interpretation since it uses to the lower side of the confidence interval of the mean portfolio return and divides

it by the mean portfolio turnover. Note thereby that a higher turnover leads to lower critical trading costs. This especially holds for strategies, which imply high turnover such as all momentum strategies.

Most important, the results from the approximate formula are always close to the exact formula (the results for the exact formula are not reported but are available on request).

Table A.1: Portfolio sorts for factor construction

		Six size-BE/ME portfolios		
		BE/ME		
		low	medium	high
size	small	S/L	S/M	S/H
	big	B/L	B/M	B/H

Note: This table illustrates the sorting procedure which is used to create six size-BE/ME portfolios which are the building blocks of the SMB and HML. All stocks are divided into two size groups by their market value (small (S) and big (B)). Simultaneously all stocks are also divided into three BE/ME groups (low (L), medium (M) and high (H)). For a discussion of the breakpoints see Schmidt et al. (2017).

Highlights

- We study the link between the profitability of momentum strategies and firm size, covering 14 stock markets across the globe.
- International momentum profitability is markedly higher in medium-size than in big stocks.
- Momentum premia are considerably diminished by trading costs.
- Momentum premia especially for medium-sized stocks still remain economically and statistically significant in most equity markets worldwide after adjusting for trading costs.